

SAMRUDHI BADAKH

Computer Science Undergraduate | COEP Technological University
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PROFESSIONAL SUMMARY

Motivated Computer Science undergraduate at COEP Technological University with hands-on experience spanning embedded systems and web development. Contributes to a real satellite project as part of the Communication Software team, working on low-level SPI/SD card drivers for the SAM4S microcontroller in C. Also experienced in building web applications with React, Three.js, and FastAPI. Seeking software or embedded development internships to apply and expand these skills in industry.

EDUCATION

COEP Technological University – Pune, Maharashtra
Bachelor of Technology – Computer Science & Engineering
2024 – Present

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java, SQL

Embedded & Hardware: SPI Protocol, ARM Cortex-M4, SAM4S Microcontroller, Register-level Programming

Web Technologies: HTML, CSS, JavaScript, React, Three.js, FastAPI

Developer Tools: Git, GitHub, GitLab, VS Code

Operating Systems: Windows, Linux

Core Competencies: Data Structures & Algorithms, OOP, Problem Solving, Debugging

PROJECTS

SD Card Interface via SPI – SAM4S Microcontroller | University Satellite Club

- Developed a low-level SPI driver in C to interface an SD card with the SAM4S ARM Cortex-M4 microcontroller for onboard data storage in a satellite system
- Handled SPI initialization, chip-select signaling, and SD command/response protocol (CMD0, CMD8, ACMD41) to achieve reliable card initialization
- Enabled raw block read/write operations; part of the Communication Software sub-team contributing to the university's active satellite project

Face Detection & Recognition System

- Built a real-time face detection and recognition pipeline in Python using OpenCV and the face_recognition library (dlib-based deep metric learning)
- Implemented face encoding, multi-face tracking across video frames, and identity matching against a known-faces database
- Exploring CNN-based approaches to improve robustness under varying lighting, angles, and occlusion

Three.js CSAT Website

- Built an interactive 3D website for the university satellite club using Three.js and vanilla JavaScript
- Implemented custom 3D scene management, camera controls, and animations for an immersive web experience

Todo REST API – FastAPI

- Built a RESTful task management API using Python and FastAPI with full CRUD operations and automatic Swagger/OpenAPI documentation
- Implemented request validation via Pydantic models; hands-on experience with REST principles, HTTP methods, and Python async patterns

ACHIEVEMENTS & ACTIVITIES

- Qualified JEE Advanced 2024
- 97.6 percentile in JEE Mains 2024 (among top 2.4% nationwide)
- 84.33% in HSC (Class 12th)
- 97.83% in CBSE (Class 10th)
- Active member: University Satellite Club, Table Tennis Team, Basketball Team